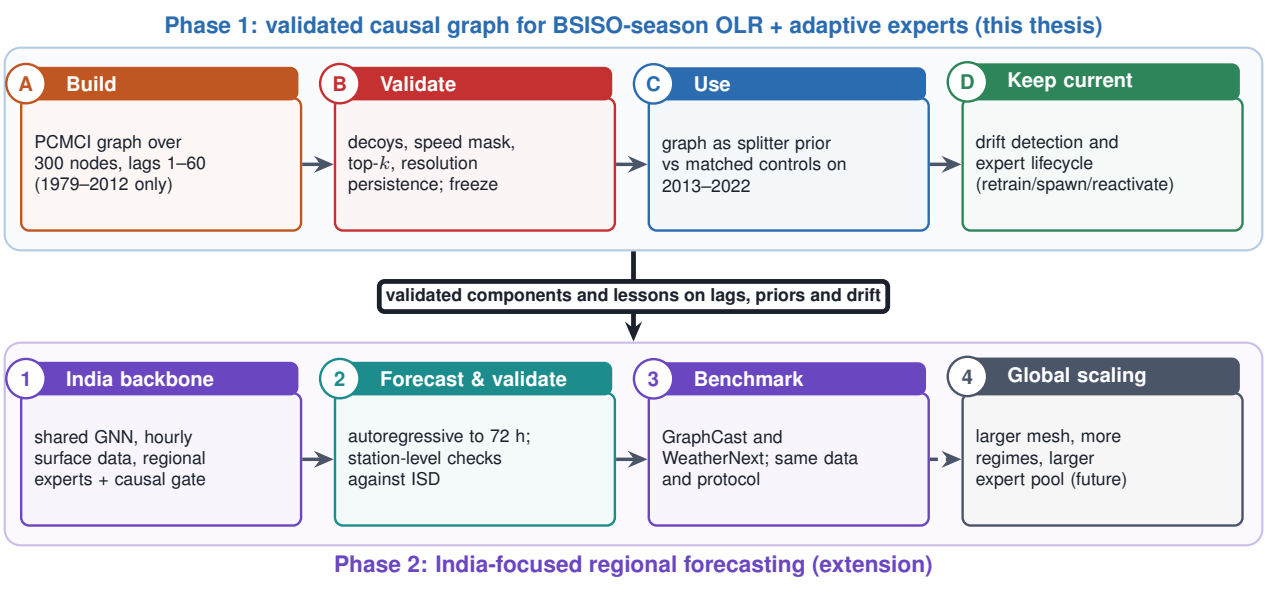
Phase 1: validated causal graph for BSISO-season OLR + adaptive experts (this thesis)
A
Build
PCMCI graph over
300 nodes, lags 1–60
(1979–2012 only)
B
Validate
decoys, speed mask,
top-k, resolution
persistence; freeze
C
Use
graph as splitter prior
vs matched controls on
2013–2022
D
Keep current
drift detection and
expert lifecycle
(retrain/spawn/reactivate)
validated components and lessons on lags, priors and drift
1
India backbone
shared GNN, hourly
surface data, regional
experts + causal gate
2
Forecast & validate
autoregressive to 72 h;
station-level checks
against ISD
3
Benchmark
GraphCast and
WeatherNext; same data
and protocol
4
Global scaling
larger mesh, more
regimes, larger
expert pool (future)
Phase 2: India-focused regional forecasting (extension)